Wilson encountered background noise that proved impossible to remove (even after they scraped bird droppings—"white noise"— from the inside of the antenna) and, with the key insights of Robert Dicke at Princeton and his students Peter Roll and David Wilkinson, together with Jim Peebles, it was ultimately realized that the antenna was picking up microwave radiation that originated with the big bang. (Important work in cosmology that set the stage for this discovery was carried out earlier by George Gamow, Ralph Alpher, and Robert Herman.) As we discuss further in later chapters, the radiation gives us an unadulterated picture of the universe when it was about 300,000 years old. That's when electrically charged particles like electrons and protons, which disrupt the motion of light beams, combined to form electrically neutral atoms, which, by and large, allow light to travel freely. Ever since, such ancient light—produced in the early stages of the universe—has traveled unimpeded, and today, suffuses all of space with microwave photons.

5. The physical phenomenon involved here, as discussed in Chapter 11, is known as *redshift*. Common atoms such as hydrogen and oxygen emit light at wavelengths that have been well documented through laboratory experiments. When such substances are constituents of galaxies that are rushing away, the light they emit is elongated, much as the siren of a police car that's racing away is also elongated, making the pitch drop. Because red is the longest wavelength of light that can be seen with the unaided eye, this stretching of light is called the redshift effect. The amount of redshift grows with increasing recessional speed, and hence by measuring the received wavelengths of light and comparing with laboratory results, the speed of distant objects can be determined. (This is actually one kind of redshift, akin to the Doppler effect. Redshifting can also be caused by gravity: photons elongate as they climb out of a gravitational field.)

6. More precisely, the mathematically inclined reader will note that a particle of mass *m*, sitting on the surface of a ball of radius R and mass density ρ, experiences an acceleration, d^2R/dt^2 given by $(4\pi/3)R^3G\rho/R^2$, and so $(1/R)\ d^2R/dt^2 = (4\pi/3)G\rho$. If we formally identify *R* with the radius of the universe, and ρ with the mass density of the universe, this is Einstein's equation for how the size of the universe evolves (assuming the absence of pressure).